

Supplementary Material: Interviews[★]

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ABSTRACT

This paper investigates the integration of machine learning (ML) techniques into car insurance premium pricing, a domain traditionally dominated by generalized linear models (GLMs). While GLMs are widely used due to their transparency, interpretability, and regulatory acceptance, advances in ML have created opportunities to improve predictive performance. The study examines whether ML models can enhance premium pricing accuracy while maintaining the level of explainability required for practical adoption in the insurance industry. Several ML algorithms, including LightGBM, XGBoost, and a multi-layer perceptron, are evaluated and compared with traditional GLMs on both claim frequency and claim severity datasets provided by a Dutch insurer. Model performance is assessed using multiple predictive accuracy metrics. To address the interpretability challenges associated with complex ML models, explainable artificial intelligence (XAI) techniques, specifically SHAP and LIME, are applied to provide insights into model predictions and feature importance. The results show that ML models consistently outperform GLMs in predictive accuracy for both frequency and severity prediction tasks. However, findings from interviews with insurance industry stakeholders reveal that concerns regarding transparency, explainability, and regulatory compliance remain significant barriers to widespread implementation. Based on these findings, the paper proposes a hybrid pricing framework in which ML models complement rather than replace GLMs. Such an approach leverages the superior predictive capabilities of ML while retaining the interpretability and trust associated with traditional actuarial methods. The study contributes to the growing body of research on AI-driven insurance pricing and supports the gradual adoption of ML models supported by robust explainability techniques.

1. Interviews

1.1. Exploratory Interview

This is the first exploratory interview between the author and Individual E. Individual D also joined the conversation. The aim of the exploratory interview was to examine:

1. *Understanding pure premium pricing and commercial pricing*
 - Explore the components of pure premium pricing and commercial pricing.
 - Examine the factors and aspects considered in these pricing processes.
2. *Application of GLMs in insurance pricing*
 - Analyze the advantages and disadvantages of using GLMs in insurance pricing.

- Investigate why GLMs are still preferred over ML techniques in current practices.

3. *Application of ML in insurance pricing*

- Analyze the advantages and disadvantages of using ML in insurance pricing.
- How explainability plays a part in insurance pricing.

During the interview, the components of pure premium pricing and commercial pricing were explored extensively. Pure premium pricing involves calculating the base premium based on risk factors such as the type of coverage, building type, province, and occupational category. This pure premium is crucial as it directly influences the final commercial pricing by incorporating additional factors. In commercial premium pricing, several additional costs are added to the base risk (pure) premium, including commissions, underwriting fees, and operational expenses. This also includes analyzing loss ratios, which are influenced by factors like the size of the company and the frequency and severity of the registered claims. The importance of understanding the reasons behind high loss ratios was stressed, such as whether claims are due to large individual claims or a higher frequency of smaller claims. This analysis helps in setting appropriate premiums that reflect the actual risk.

The insurer offers a variety of insurance policies, most of which pertain to cars. These car insurance policies are primarily managed using GLMs. For other products, pricing is more frequently based on loss data and different influencing factors. One of the primary benefits of GLMs is

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their explainability. GLMs allow for a clear understanding of how each factor affects the outcome, which is crucial for explaining model decisions to stakeholders, including regulatory bodies. It is also mentioned by Individual D that it is important that the gross of data is predicted correctly, rather than shifting too much modeling fitting towards the few outliers that exist within a claim portfolio. Individual E explained that for auto insurance, the insurer uses GLMs to evaluate numerous variables, such as the car's brand, the driver's age, and regional risk factors. They assess deviations in these variables to determine their impact on the overall risk. For instance, they have identified that certain high-value cars do not present twice the risk of cars valued at two million euros. This explanatory insight helps in setting premiums that are fair and competitive, avoiding disproportionate increases for higher-valued vehicles.

A significant part of the discussion revolves around the application of ML models, balancing predictive accuracy and explainability. The benefits of ML, such as the possibility of having lower prediction errors, are acknowledged. However, there is a concern about the black-box nature of these models and its variability, which could hinder clear explanations to clients and advisors. Customers often prefer stable premiums over potentially lower, variable premiums due to predictability in expenses, similar to fixed mortgage rates. Stability is especially important in high-value items like car insurance, where customers value the assurance that their premiums won't fluctuate significantly year-to-year. Price comparison tools like Independer² play a significant role in how customers choose their insurance. When faced with slightly varying premiums, many customers prefer the stability of a higher, but more predictable premium. This lack of explainability is a significant drawback in insurance pricing with ML, where understanding the rationale behind premium calculations can be essential.

The insurer emphasizes the importance of explaining premium calculations to customers, especially when premiums increase significantly. Tools and dashboards can help illustrate the factors contributing to premium changes. However, complex models like those generated by ML pose challenges in explainability compared to GLMs. These models allow insurers to explain price changes based on specific, understandable factors. However, there is interest in exploring ML for its potential to improve pricing accuracy. While the insurer is open to innovation, there is also a cautious approach to implementation of ML in insurance pricing, considering customer and advisor expectations. Transitioning to ML models involves significant initial costs and complexities. These include training the models, validating their results, and ensuring they are robust enough for real-world application. Additionally, models must be reproducible and explainable to satisfy both regulatory bodies and customers. The insurer highlights specific cases like Porsche insurance, where the premium might increase significantly due

to higher risks associated with certain car models. These decisions need to be explained clearly to customers to maintain trust. The idea of a pilot project for a new product using ML is proposed, allowing for monitoring results and integrating lessons learned before broader implementation. Both parties agree on the necessity of innovation in risk pricing. However, the implementation must be careful and transparent to keep clients and advisors well-informed and engaged. A pilot project for a new product seems the best way to test ML techniques and evaluate their effectiveness, providing a pathway to potentially broader application if successful.

1.2. Final Interviews

This appendix section handles the various semi-structured interviews conducted with respective stakeholders to identify possibilities of integrating ML while trying to preserve explainability and transparency of these models. Here, all the interviews are summarized. The interview was divided into the following main topics:

1. *Technical Background.* Basic information as education, previous work experience, and the current job description of the interviewee are explored in this part of the interview. Understanding the interviewee's technical background provides valuable context for their responses and insights. It allows to assess the level of expertise and experience they bring to the discussion, which is crucial for evaluating their perspective. Questions involve:
 - Could you share details about your educational background? (will not be disclosed in the paper due to anonymization)
 - What professional experience do you have? (will not be disclosed in the paper due to anonymization)
 - Can you describe your role and responsibilities within the team? (will not be disclosed in the paper due to anonymization)
2. *Current Practices.* The current practices and methodologies employed by the specific department in the insurance company's pricing process are explored in this phase of the interview. This includes the tools, models, and strategies they currently use. Understanding the current practices within an insurance company is essential to evaluate the potential impact and feasibility of integrating ML algorithms.
 - How many years of experience do you have working with GLMs?
 - Could you walk me through the entire model development process?
3. *Model Validation.* This process is crucial for ensuring the accuracy and reliability of predictive models. Model validation involves a variety of techniques and metrics to assess how well a model performs on unseen data. Each method provides different insights into the model's performance and error distribution.

²Independer.nl N.V. (operating under the name Independer) is a Dutch comparison site by DPG Media for financial products, healthcare institutions, and energy.

Evaluating the performance of ML models compared to traditional methods such as GLMs is essential for understanding the potential benefits and drawbacks of each approach. By comparing these models, one can determine whether the results are compelling enough to make a transition to or inclusion of ML techniques in the modeling process. Proper validation ensures that models are robust, generalizable, and capable of making accurate predictions in real-world scenarios.

- What methods do you use to validate the current models (e.g., MAE, RMSE, MSE, others)?
 - How do you interpret the results of your validation metrics in a business context?
 - How do you evaluate the performance of ML compared to traditional methods such as GLMs?
 - Are the ML results compelling enough to consider using this technique?
 - What can you say about the APP plots for ML compared to GLMs? Are they comparable?
 - In your opinion, what could be a potential drawback of applying ML techniques?
4. *Importance of Model Explainability.* This topic seeks to uncover how crucial it is for the team to be able to explain and understand the models they use. It explores the role of explainability in building trust and confidence in the models, how it influences the adoption of new methodologies, and any specific instances where the ability to thoroughly explain a model's output was particularly beneficial.
- Can you provide examples where the explainability of GLMs was advantageous? What did you need at that time?
 - How do you view the importance of explainability within the insurance sector? Is there room to reduce this for better pure premium pricing precision?
5. *XAI Outcome Analysis.* Focusing on evaluating the global and local outcomes provided by XAI techniques used. The aim is to gather detailed feedback so these explanations can be compared to the traditional factor-based explanations provided by GLMs, assessing the usefulness of these explanations for ML techniques. The goal is to understand how well these XAI techniques meet the department's needs for transparency and whether they facilitate a better understanding of ML models.
- (Global) How can the global XAI outcomes be compared to traditional factor-based explanations of GLMs?
 - (Global) Do the outcomes align with your experience and expertise?
 - (Global) What is your perspective on tools like SHAP, and do you trust the results they generate? Can you use them to achieve what you would typically do with GLMs to some extent?

- (Local) How can the local XAI outcomes be compared to traditional factor-based explanations of GLMs?
 - (Local) Do the outcomes align with your experience and expertise?
6. *Comparison GLMs and ML Explainability.* Centers around identifying the strengths and weaknesses of each approach and understanding the interviewee's perspective on the feasibility of adopting ML models in light of their explainability.
- Do you see specific advantages or disadvantages in using SHAP and LIME compared to GLM explanations?
 - Based on the evaluation, do you see a future where ML models could replace GLMs in your pricing process?
 - What additional information or results would you need to make an informed decision about the adoption of ML models?
7. *Views on AI Adoption.* Looking into the overall potential and readiness for AI adoption within the current infrastructure and workflow.
- What potential challenges do you foresee in integrating ML algorithms into the current pricing process?
 - Are there infrastructural or resource limitations that could impact the adoption of ML models?
 - How well do you think your team is prepared to transition from GLM to ML models in terms of skills and knowledge?
 - What kind of training or resources would be necessary to facilitate this transition?
8. *Feedback and Suggestions.* This ensures that no steps are overlooked in the process and that any critical points vital to the project's outcome are included.
- Do you have any feedback on the ML models and their explanation that can help improve their integration and acceptance?
 - Are there any other aspects or factors that you think should be considered when evaluating the transition from GLMs to ML models?

1.2.1. Interview #1: Individual A

Individual A has worked on a variety of projects involving dashboarding and predictive modeling, particularly for insurance products. Individual A's role has primarily focused on developing ML models for premium pricing, although Individual A acknowledges that GLMs remain the industry standard today.

The model development process at an insurer made was validated, looking into a general insurance company involving several stakeholders, including a commercial risk committee that evaluates the models for sufficient risk margins, external actuaries who validate the models against

regulatory standards, and a product management team that deploys the models. This process ensures that the models are robust, compliant, and commercially viable. In some larger organizations, according to Individual A, this is done by internal actuaries. They develop the models, which are then reviewed by commercial departments. Therefore, the model could use some level of revision to make it more generalized.

To validate the models, various metrics such as MAE, RMSE, MSE, and tuning time are employed. MAE was highlighted as particularly important because it focuses on the bulk of the data around the mean, reducing the impact of outliers. Individual A agrees that if this emphasis aligns with some insurers' priorities, which value the accuracy of the majority of predictions over the extremes. Individual A interprets validation results by comparing predicted values against actual values, focusing on the mean and distribution across different segments. Individual A notes that ML models, such as neural networks and tree-based models, show varying performance, with neural networks excelling in certain areas despite longer tuning times. The performance of ML models is evaluated through metrics and plots that compare actual versus predicted values, called APP plots. ML models tend to show more accurate predictions for segments with larger variances compared to GLMs. This detailed analysis helps in understanding the strengths and weaknesses of different models. When comparing APP plots, Individual A finds that ML models generally offer more precise predictions, especially in categories with significant variances. This increased accuracy is crucial for fine-tuning insurance premiums. Potential drawbacks of ML include the longer tuning times required for some models and the need for extensive validation to ensure accuracy. These factors must be weighed against the benefits when considering the adoption of ML models. Explainability is crucial for ensuring models are free from errors and for validating the importance of features. Traditional GLMs allow for easy back-calculation of factors, which some actuaries prefer. In the insurance sector, explainability helps in validating models internally and can be vital for customer-facing applications like fraud detection. However, for premium determination, Individual A believes that current explainability tools for ML, such as SHAP, provide sufficient insights.

The usage of SHAP and LIME values to explain feature importance in ML models is also discussed. The interview finds that these outcomes align well with traditional factor-based explanations from GLMs, offering a robust and deterministic method for model validation. Local XAI outcomes, which provide insights on individual predictions, are particularly useful for explaining differences in premiums for similar cases. SHAP is preferred for its robustness and deterministic nature, as LIME's variability makes it less reliable. Individual A believes that the visual and interpretative clarity provided by SHAP and other XAI tools can match the explanatory power of GLMs, making them viable for use in insurance pricing.

Looking ahead, Individual A sees the potential for ML models to replace GLMs in the pricing process, provided

that explainability and validation concerns are adequately addressed. He emphasizes the importance of continued validation and stakeholder acceptance for a successful transition. To make an informed decision about adopting ML models, additional comprehensive validation results and detailed insights into feature impacts are necessary. This information helps build confidence in the accuracy and reliability of ML models. Integrating ML algorithms into existing pricing processes poses several challenges, including meeting regulatory standards and ensuring model transparency. Infrastructural and resource limitations, such as the need for specialized knowledge and computational power, can also impact adoption. Individual A believes that teams need to be adequately trained in ML techniques and tools like SHAP to effectively transition from GLMs to ML models. Ongoing training and access to necessary resources are crucial for facilitating this transition. For better integration of ML models, Individual A suggests enhancing explainability tools and streamlining validation processes. Developing a more generic model to accommodate different types of insurers, from small to large companies, is also recommended ensuring wider applicability and acceptance. Overall, Individual A's insights highlight the potential for ML models to revolutionize insurance pricing, provided that challenges around explainability and validation are addressed adequately.

1.2.2. Interview #2: Individual B

Individual B has been involved in various facets of non-life insurance, including pricing. Individual B's experience has given elaborate understanding of both traditional actuarial models and modern ML techniques. In Individual B's current role, Individual B focuses on model validation and advice, ensuring that the models used within Individual B's organization are robust, compliant, and commercially viable. Individual B emphasizes the importance of understanding the practical implications of model outputs and the necessity of aligning them with regulatory standards and business objectives.

Individual B acknowledges the prevalence of GLMs in the industry, noting that while ML models offer promising results, GLMs remain the standard due to their explainability and regulatory acceptance. Individual B acknowledges the model development process in various organizations involves multiple stakeholders, which in case of the insurer a commercial risk committee and external actuaries, ensuring that models are thoroughly vetted and validated before deployment.

The organizational structure and stakeholder involvement was also discussed. Individual B noted that various insurers operate with a dual approach: one following some more traditional methods such as fixed formulas and another incorporating more advanced (GLM) models. This dual approach allows to benefit from both worlds, with traditional methods for regulatory compliance and newer models for internal analysis and customer-specific adjustments. Individual B highlights the importance of focusing on validation metrics, aligning with insurers' priorities for

accurate predictions around the mean. When interpreting these validation results, the author showed comparisons between the predicted values against actual values, focusing on the mean and distribution across different segments. From the findings presented, Individual B concludes that ML models offer more precise predictions. This increased accuracy could be crucial for fine-tuning insurance premiums. However, he also acknowledges potential drawbacks of ML, including longer tuning times and the need for extensive validation. Explainability is the key concern for Individual B, particularly in the context of risk management. Individual B believes that understanding model outputs and their implications is crucial for making informed decisions. Traditional GLMs offer ease of back-calculating factors, which some actuaries prefer. The author used the retrieved SHAP values to explain feature importance in ML models. Individual B confirms that these outcomes align well with traditional factor-based explanations from GLMs, although GLMs provide more detailed explanations. Additionally, it is challenging to identify relationships between variables using SHAP.

Looking ahead, Individual B sees the potential for ML models to replace GLMs in the pricing process, provided that explainability and validation concerns are adequately addressed. Individual B emphasizes the importance of continued validation and stakeholder acceptance for a successful transition. To make an informed decision about adopting ML models, additional comprehensive validation results and detailed insights into feature impacts are necessary. Integrating ML algorithms into existing pricing processes poses several challenges, including meeting regulatory standards and ensuring model transparency. Individual B highlights infrastructural and resource limitations, such as the need for specialized knowledge and computational power, as potential barriers to adoption. At a lot of insurers, this is deemed insufficient for now. For better integration of ML models, Individual B suggests enhancing explainability tools and streamlining validation processes. Individual B also recommends developing a more generic model to accommodate different types of insurers, ensuring wider applicability and acceptance.

1.2.3. Interview #3: Individual C

The interview with Individual C delves into the insurance pricing models, focusing on the comparison between GLMs and ML, and the practical challenges and potential benefits of adopting new technologies in the industry. Individual C's work primarily involves Solvency II, balance and capital management, and actuarial functions.

The conversation started discussing the organizational structure of the actuarial modeling process at the insurance company, where Individual C explains that the Data Management team creates models as the first line of defense. The team combines inputs from various departments, including the commercial pricing department, to develop the pricing models used by the company. These models are then reviewed by the actuarial function, which Individual C is a

part of, representing the second line. The role of the second line is to review and audit the models created by the first line whether these meet regulatory and internal standards, ensuring an unbiased review process. The third line of defense is typically the internal and external audit functions. In the case of the insurer, the internal actuarial team performs annual norm validation checks on the models used. However, this internal team is limited in size and scope. The third line focuses on compliance and high-level review to ensure that the first and second lines are functioning correctly.

The interview continued by delving into the comparison between GLM and ML models. Individual C emphasizes that while GLM models are currently more explainable and easier to validate, ML models, despite their potential for higher accuracy, present significant challenges due to their black-box nature. This lack of transparency and explainability makes it difficult for insurers to fully trust and adopt ML models without robust validation methods in place. The validation process for GLM models at the insurance company involves deep reviews of code and data, a thoroughness that Individual C notes might be specific to the insurance company and not as common in other insurance companies. The importance of feature importance plots and XAI methods in making ML models more explainable is also discussed, where Individual C likes the factor-based GLMs due to its thoroughness. However, Individual C acknowledges that while these methods can partially explain ML models, they do not offer an in-depth analysis of the relationships between individual variables. He also addresses the difficulties in adopting ML models due to these transparency issues.

The interview continues discussing the potential issues of data limitations and the risk of mispricing in segments with insufficient data. Individual C suggests that ML could be used alongside GLMs rather than replacing them entirely. This hybrid approach could combine the strengths of both models, leveraging ML's accuracy while maintaining the transparency and explainability of GLMs. Individual C acknowledges that while ML can offer potential in more accurate pricing algorithms, the transparency and ease of validation inherent in GLM models currently make them more suitable for insurance pricing. Individual C points out that some insurers, especially smaller ones, are still not even using GLMs, indicating a significant variance in the adoption of pricing models across the industry.

Afterwards, a discussion started talking about the possibility of using ML models to enhance the feature selection process for GLMs. This would allow insurers to benefit from the strengths of both approaches, facilitating a gradual integration of ML into existing pricing processes. He emphasizes the need for additional information and results to make ML models adoptable and suggests involving ML experts for proper validation, as this is not Individual C's primary field of work.

Towards the end of the interview, the conversation touches on the future adoption of ML in insurance. Individual C notes that while ML could become more prevalent, it would face diminishing returns after a certain point compared to the

significant leap from judgment-based methods to GLMs. He discusses the competitive aspect, where insurers with better models can achieve better pricing, but this also raises the risk of anti-selection, where less advanced insurers might end up with worse risks. Advancements in other markets were discussed, particularly in the UK, where dynamic pricing and ML adoption might be more advanced due to different market dynamics and competitive pressures. This global perspective provides additional context to the ongoing discussion about model adoption and implementation.

Individual C's interview underscores the current state and future potential of ML in insurance pricing. While ML offers improved accuracy, its lack of explainability, even with XAI techniques, and the robust validation processes required pose significant challenges. The discussion highlights and brings up the possibility of combining ML with GLMs to leverage both their respective strengths. The necessity for a gradual, well-validated adoption of new models in the insurance industry is also discussed.

1.2.4. Interview #4: Individual D

Individual D is deeply involved in product development, portfolio management, and pricing strategies, focusing on creating and validating pricing models that align with the company's financial goals and regulatory requirements.

The insurer transitioned to GLMs around 2018, replacing simpler, older methods that relied on fixed formulas. The shift to GLMs was driven by the need for more accurate risk assessments, particularly for pricing models with variables like postcode factors. Back then, GLMs were considered as they offered several advantages, including enhanced explainability, which allows for clear, interpretable relationships between inputs and outputs. This transparency is crucial in the insurance industry, where stakeholders need to understand the rationale behind premium calculations. The transition involved refining broad regional groupings into more precise estimations using ML techniques. The implementation of GLMs required extensive validation, in which the Commercial Risk Management Committee played a role as well. This committee, comprising product management, directorial staff, and other key stakeholders, evaluates the models' impact on the company's profitability and market competitiveness. The department ensures that the models are aligned with company goals, while providing accurate and actionable insights. The committee's feedback was important to the models' refinement and validation. If necessary, the models undergo further adjustments before being finalized.

Explainability plays a crucial role in the acceptance and implementation of these models. Individual D provided examples where the explainability of GLMs was particularly advantageous. For instance, when discrepancies arise between expected and actual premium calculations, GLMs allow the team to trace back and clearly explain these differences. This capability is essential for internal validation and addressing customer inquiries about their premiums. The ability to provide detailed explanations builds trust

with stakeholders and ensures compliance with regulatory standards.

The interview explored the performance of various ML models, including the tree-based models LightGBM and XGBoost, and neural networks. Validation methods such as cross-validation and APP plots were used to ensure models were correctly handling the available data. This process helped maintain model accuracy and reliability, which is crucial for their acceptance and practical application within the organization. While Individual D recognizes that ML models can capture more complex patterns and potentially improve prediction accuracy, they often lack the transparency required for regulatory compliance and stakeholder trust. Individual D noted that ML models function as black-boxes, making it difficult to justify their predictions. Despite this, the potential benefits of ML models, particularly in terms of performance improvements, are recognized. However, any transition to more complex models would require significant advancements in explainability and integration with existing systems. While ML models showed potential, GLMs remained preferred due to their explainability and reliability. This preference is critical in contexts where the consequences of model predictions can have significant financial implications. Significant change will only occur when organizations such as the consultancy firm advise them to do so.

Comparing ML models with GLMs, Individual D discussed the use of tools like SHAP to interpret ML model predictions. SHAP values provide insights into the importance of each feature, aiding in the explainability of ML models. However, there are still concerns about fully replacing GLMs with ML models due to the latter's inherent complexity. The current preference for GLMs stems from their balance of accuracy and transparency, which ML models have yet to consistently achieve.

The discussion also covered the challenges of integrating ML algorithms into the current pricing process. This integration faces several obstacles, including the need for infrastructural upgrades and potential resource limitations. Additionally, according to Individual D, transitioning from GLMs to ML models would require extensive training and resources to equip the team with the necessary skills and knowledge. This includes understanding new validation techniques and interpreting complex model outputs. When evaluating the transition from GLMs to ML models, it is essential to consider the broader impact on operational efficiency, regulatory compliance, and stakeholder trust. The decision should be based on a thorough cost-benefit analysis, weighing the potential gains in predictive accuracy against the practical challenges of implementation.

1.2.5. Interview #5: Individual E

Throughout the conversation with Individual E, there is a significant emphasis on the comparison between traditional GLMs and more advanced ML techniques for determining insurance premiums. The core of the discussion revolves around the potential benefits and challenges associated with

transitioning from GLMs to ML models within the insurance industry, particularly in pricing strategies.

One of the central themes of the interviews is the performance and potential of ML models in improving the accuracy of insurance pricing. Individual E discusses how ML models, with their capacity to handle more extensive and complex datasets, have the potential to offer more precise predictions compared to GLMs. However, it is noted that the current GLMs in use still perform adequately, and the margin of improvement with ML models, while present, is relatively small. This leads to a nuanced consideration of whether the transition to ML is warranted at this stage, especially given the operational and interpretative challenges it presents.

A significant concern raised during the interviews is the issue of explainability. GLMs, while more straightforward and less sophisticated than ML models, offer a level of transparency that is crucial in the insurance sector. The ability to explain how premiums are calculated is vital, particularly when dealing with stakeholders such as customers and insurance advisors. Individual E highlights that the predictability and consistency offered by GLMs are valued by these stakeholders, who often need to justify premium differences between similar cases. The challenge with ML models is that, while they might offer slight improvements in accuracy, they can be seen as black-boxes, making it harder to interpret and explain the results in a clear and understandable way.

The interview also delves into the importance of certain features within the models used for predicting insurance claims costs. Individual E points out that variables such as the insured amount, the year of damage (referred to in Dutch as "schadejaar"), and the age of the vehicle could play a significant role in both GLM and ML models. These features can help in determining the likelihood and cost of claims. The discussion suggests that while ML models can potentially incorporate more variables and provide a more nuanced analysis, the traditional features still hold substantial predictive power in GLMs.

Individual E also touches on the practical challenges of implementing ML models in an operational setting. One of the key challenges is maintaining predictability and consistency in pricing, which is especially important when dealing with insurance advisors who expect similar premiums for similar risks. The consistency provided by GLMs ensures that advisors can offer reliable quotes to clients without facing significant price fluctuations, which could undermine client trust. Individual E emphasizes the need for robustness in pricing models, advocating for models that not only provide accurate predictions but also maintain a level of transparency and predictability over time.

The conversation briefly explores the future of car insurance pricing, with a mention of telematics as a potential area for development. Telematics, which involves using data on driving behavior to set premiums, could allow for more personalized and potentially more accurate insurance pricing. Individual E acknowledges that while current models focus heavily on the characteristics of the car (such as its

age and value), there is a growing need to consider factors related to the driver and their behavior. This shift could lead to more sophisticated pricing models that better reflect the actual risk. However, questions arise regarding the practicality of this approach. Mobile phones are often considered insufficient, and the challenge of asking clients to pay for their telematics tracking device remains a complex issue.

Additionally, the interviews handled the internal processes related to model validation and approval within the company. Individual E discusses the roles of various teams, including data management and internal audit, in ensuring that pricing models are robust, compliant with regulations, and effectively integrated into business operations. This process involves multiple layers of validation, where models are reviewed and adjusted before being implemented. The discussion suggests a well-structured approach to model governance, ensuring that any pricing model, whether GLM or ML, meets the company's standards for accuracy and reliability.